INTRODUCTION

The **Online Project Submission Portal** is a web-based academic management system designed to simplify and digitize the entire project submission and evaluation process in educational institutions. The system acts as a centralized platform where students, project guides, Heads of Departments (HODs), and administrators can efficiently manage academic projects in a structured and secure manner.

Traditionally, project submission and evaluation are handled manually, which often leads to paperwork, delays, data mismanagement, and lack of transparency. This portal eliminates these issues by providing an automated and user-friendly digital environment for handling all project-related activities.

Using the portal, students can upload project reports, source code files, screenshots, and other required documents online. Faculty members and project guides can review submissions, provide feedback, verify project details, and evaluate projects directly through the system. The platform also maintains proper records of submitted projects for future reference.

An additional feature of the system is the **Public Project Gallery**, where approved projects can be displayed. This allows students to explore previous academic projects for learning, research, and inspiration purposes.

The system enhances communication between students and faculty members, ensures transparency in the evaluation process, and creates a well-organized repository of academic projects within the institution.

EXISTING SYSTEM

The traditional project submission and management system used in many educational institutions is mostly manual, paper-based, and time-consuming. Students, faculty members, and administrators face several difficulties due to the absence of a centralized and digital platform for handling academic projects.

Manual Project Submission: Students submit project reports, source code, and related documents through emails, CDs, pendrives, or printed copies, which makes the submission process slow and unorganized.

- **Lack of Centralized Storage:** Project files and records are stored in different locations, making it difficult to manage, access, and maintain project data efficiently.
- **Difficulty in Maintaining Project History:** Tracking previous submissions, updates, and evaluation records becomes complicated due to manual record management.
- **Delayed Communication:** Communication between students and project guides is often slow, causing delays in approvals, feedback, and project evaluation.
- **Risk of Data Loss:** Physical storage devices such as CDs and pendrives can be damaged, misplaced, or corrupted, leading to loss of important project files.
- **No Standard Evaluation Process:** The project review and evaluation process is handled manually, which may create inconsistencies and reduce transparency.
- **Limited Access to Previous Projects:** Students cannot easily access old academic projects for learning, research, and reference purposes.
- **Increased Paperwork and Administrative Workload:** Manual handling of project records increases paperwork and workload for faculty members and administrators.
- **Difficulty in Tracking Project Progress:** Monitoring submission status, project approvals, and feedback becomes difficult without a proper online system.
- **Limited Security and Backup Facilities:** The traditional system does not provide proper security, backup, or recovery mechanisms for important project data.

Due to these limitations, the existing project management system fails to meet the requirements of modern educational institutions. Therefore, there is a strong need for a smart, automated, and centralized **Online Project Submission Portal** that improves efficiency, transparency, accessibility, and secure management of academic projects.

PROPOSED FUNCTIONALITY

The proposed **Online Project Submission Portal** is a web-based system designed to simplify and automate the management of academic projects in educational institutions. The portal brings all project-related activities into a single digital platform to improve efficiency, transparency, accessibility, and secure data management for students, faculty members, and administrators.

1. Student Registration and Login

Students can securely register and log in to the portal using their credentials. Features include:

- Secure authentication system
- Easy profile management
- Access to personal project dashboard

2. Team Formation and Guide Assignment

The portal allows students to create project teams and assign guides for project supervision. Features include:

- Team creation and management
- Guide allocation by department
- Approval and monitoring by faculty members

3. Project Submission with File Upload

Students can upload project-related documents directly through the portal. Users can:

- Upload project reports, source code, and screenshots
- Submit multiple project files securely
- Update project documents before deadlines

4. Project Evaluation and Feedback

Faculty members can review and evaluate projects online through the system. Features include:

• Online project verification and approval

• Feedback and remarks from guides

• Transparent evaluation process

5. Public Project Gallery

Approved projects can be displayed publicly for learning and reference purposes. Features include:

• Access to previous academic projects

• Search and view project details

• Learning support for students and researchers

6. Role-Based Dashboard Access

The portal provides different dashboards for each user role:

• Students can manage submissions and view feedback

• Guides can review and evaluate assigned projects

• HODs and Clerks can monitor project activities

• Administrators can manage users and system operations

7. Secure Data Storage with MySQL

All project and user data is stored securely in a MySQL database with:

• Secure project file management

• Role-based access control

• Backup and recovery support for data safety

8. Search & Filter Options

The portal provides search and filtering facilities for easy data access. Users can:

• Search projects by title, technology, or department

• Filter projects by status or academic year
• Access previous project records quickly

9. Report Generation

The system can generate different reports for administrative purposes. Features include:
• Project submission reports
• Department-wise project summaries
• Student and guide activity reports

10. Security and Transparency

The portal includes important security and monitoring features such as:
• Secure login and authentication
• Role-based permissions and access control
• Transparent project review and tracking system

TOOLS, PROGRAMMING LANGUAGES, AND DATABASE USED

1. Hardware Requirements

• **Operating System:** Windows 7 and above

• **RAM:** 8 GB

• **Storage:** SSD – 512 GB, HDD – 1 TB

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2. Software Tools

• **Visual Studio Code (VS Code):**

Used for frontend and backend development of the web application.

• **Python Flask Framework:**

Used for developing backend functionalities and handling server-side operations.

• **SQLite / MySQL Server:**

Used for storing and managing project and user-related data securely.

• **Git & GitHub:**

Used for version control and project management.

• **Web Browser:**

Google Chrome / Mozilla Firefox used for testing and running the web application.

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3. Programming Languages

Frontend

• **HTML5 (HyperText Markup Language):**

Used to create the structure of web pages such as login pages, dashboards, project submission forms, and gallery pages.

• **CSS3 (Cascading Style Sheets):**

Used for styling the application and creating responsive, user-friendly interfaces across different devices.

• **JavaScript:**

Used to add interactivity, form validation, and dynamic updates for a better user experience.

• **Bootstrap 5:**

Used for developing responsive layouts and modern UI components quickly and efficiently.

Backend

• **Python Flask:**

Used for building server-side logic and handling core functionalities such as user authentication, project submission, evaluation, and report generation.

• **Flask-Session:**

Used for secure session management and maintaining user login sessions.

• **Bcrypt:**

Used for password encryption and authentication security within the system.

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4. Database – SQLite / MySQL

SQLite and MySQL are used as Relational Database Management Systems (RDBMS) for securely storing and managing all important project-related data.

Key Components Stored Include:

• **Student Information:**

Names, email IDs, department details, login credentials, and academic information.

• **Project Information:**

Project titles, descriptions, technologies used, project files, and submission details.

• **Guide and Faculty Details:**

Guide allocation records, feedback, remarks, and evaluation details.

• **Team Information:**

Project team members, assigned guides, and team management records.

• **Project Evaluation Records:**

Marks, feedback, approval status, and review history.

• **System Activity Data:**

User login records, submission activities, and administrative reports.

ENTITY RELATIONSHIP DIAGRAM (ERD) :

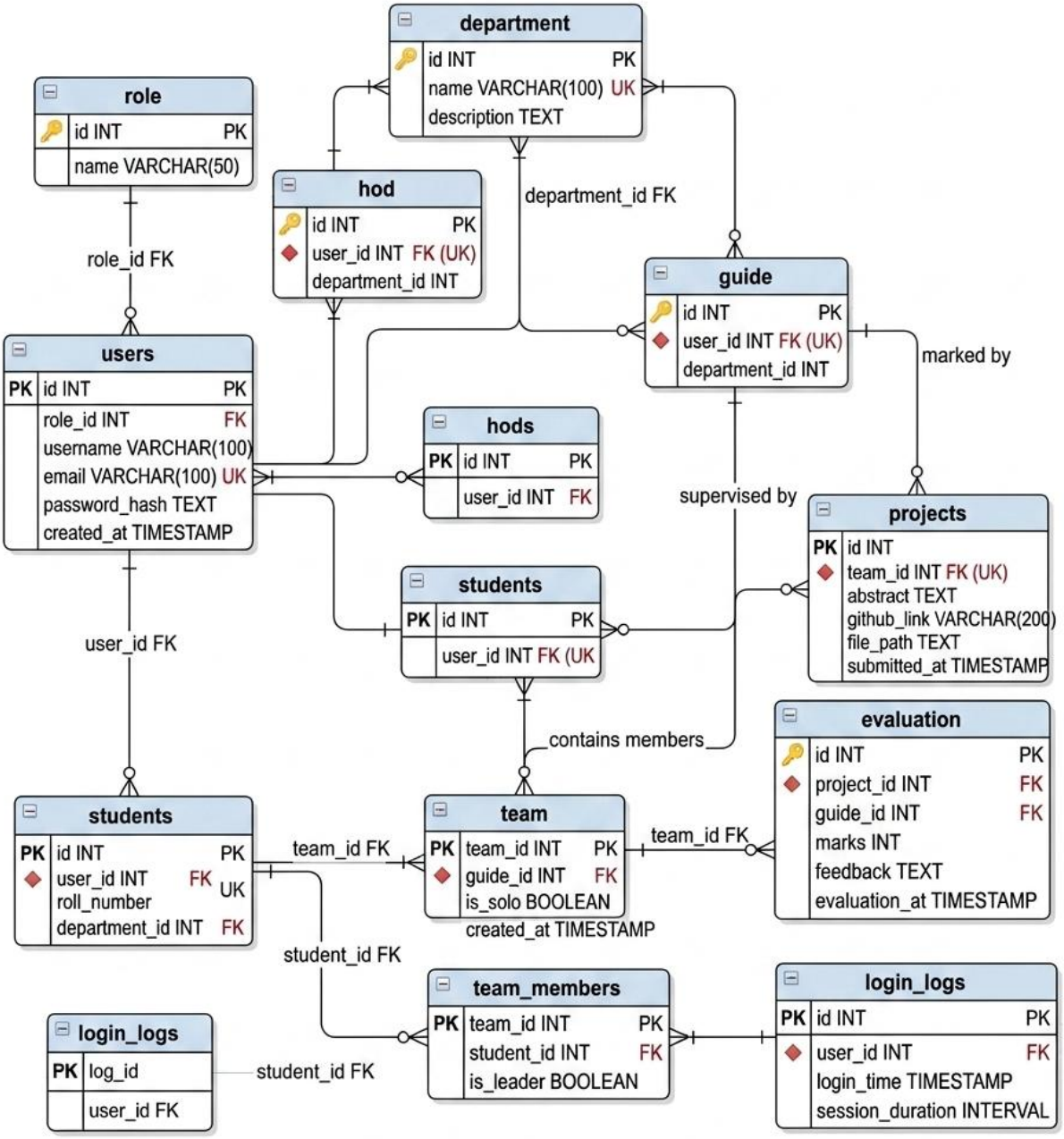


fig. 1 Entity Relationship Diagram

DATA FLOW DIAGRAM (DFD) :

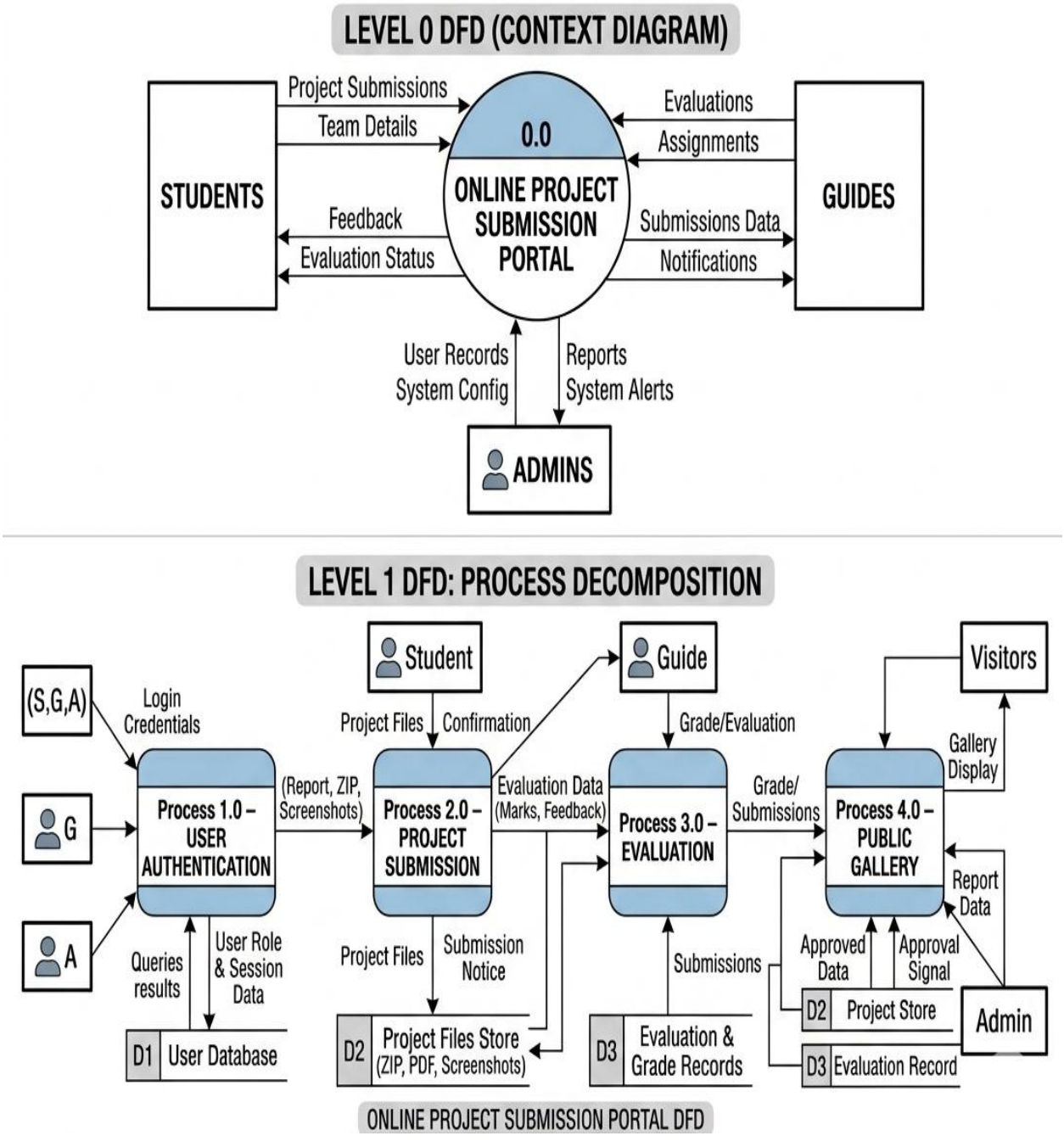


Fig. 2 Data Flow Diagram

DATA DICTIONARY (DATABASE STRUCTURE)

Tables:

1. Departments:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	NO	UNI	NULL	
description	text	YES		NULL	

2. Evaluations :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

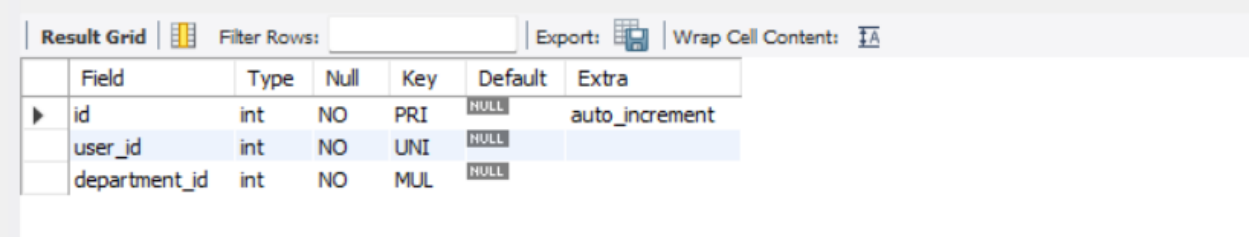
Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
project_id	int	NO	UNI	NULL	
guide_id	int	NO	MUL	NULL	
marks	decimal(5,2)	YES		NULL	
feedback	text	YES		NULL	
evaluated_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TI...

3. Guides:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	UNI	NULL	
department_id	int	NO	MUL	NULL	

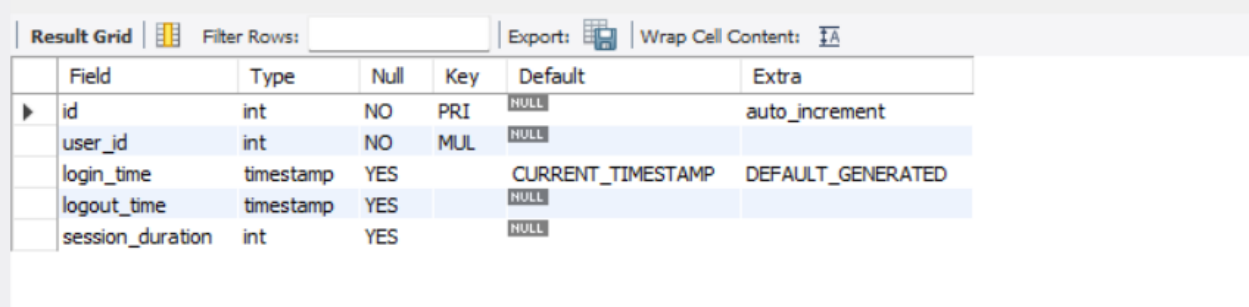
4. Hods :



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	UNI	NULL	
department_id	int	NO	MUL	NULL	

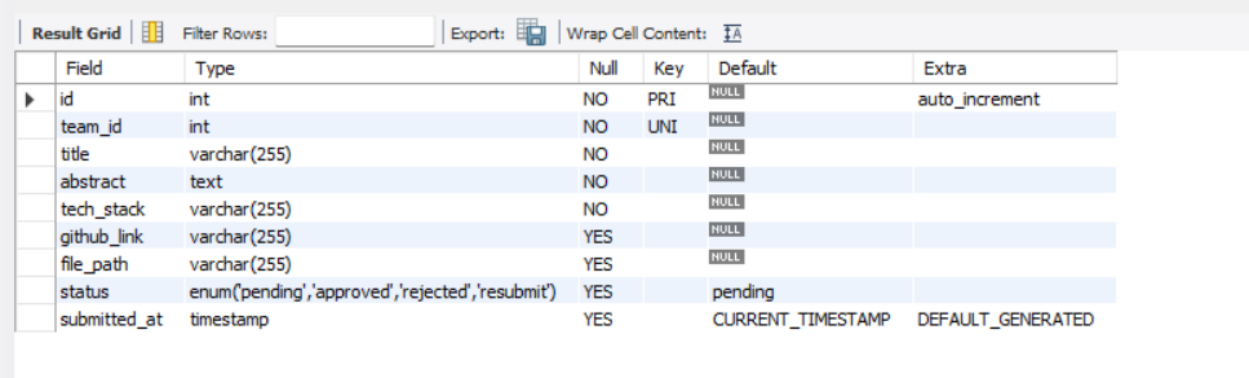
5. Login_logs :



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	MUL	NULL	
login_time	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
logout_time	timestamp	YES		NULL	
session_duration	int	YES		NULL	

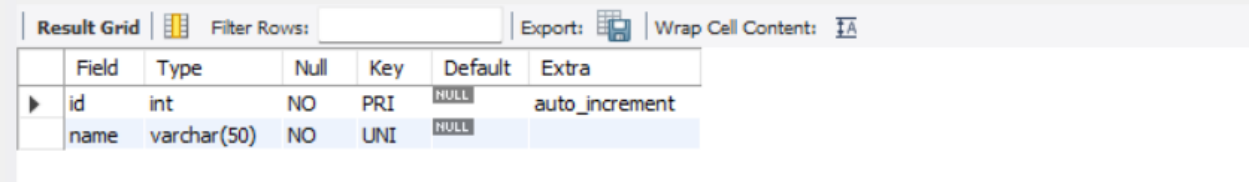
6. Projects :



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
team_id	int	NO	UNI	NULL	
title	varchar(255)	NO		NULL	
abstract	text	NO		NULL	
tech_stack	varchar(255)	NO		NULL	
github_link	varchar(255)	YES		NULL	
file_path	varchar(255)	YES		NULL	
status	enum('pending','approved','rejected','resubmit')	YES		pending	
submitted_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

7. Roles :



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(50)	NO	UNI	NULL	

8. Students :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	UNI	NULL	
roll_number	varchar(50)	NO	UNI	NULL	
department_id	int	NO	MUL	NULL	

9. Team_members :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

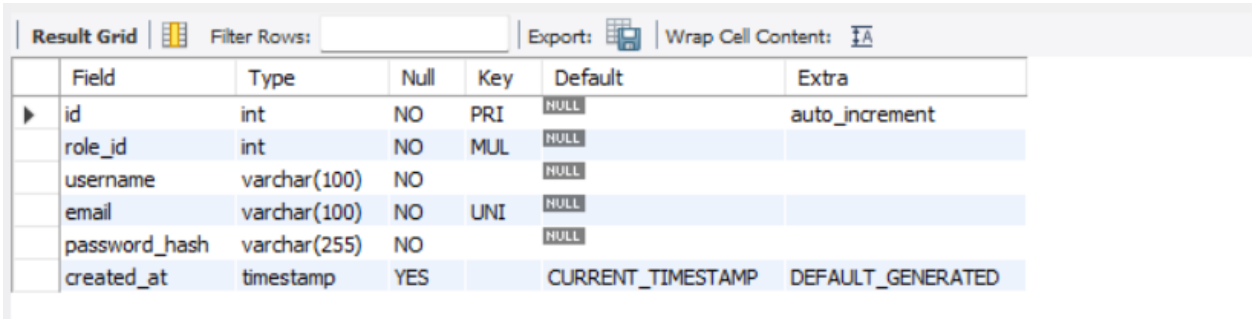
Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
team_id	int	NO	MUL	NULL	
student_id	int	NO	UNI	NULL	
is_leader	tinyint(1)	NO		0	

10.Teams :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	NO	UNI	NULL	
guide_id	int	YES	MUL	NULL	
is_solo	tinyint(1)	NO		0	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

11.Users:

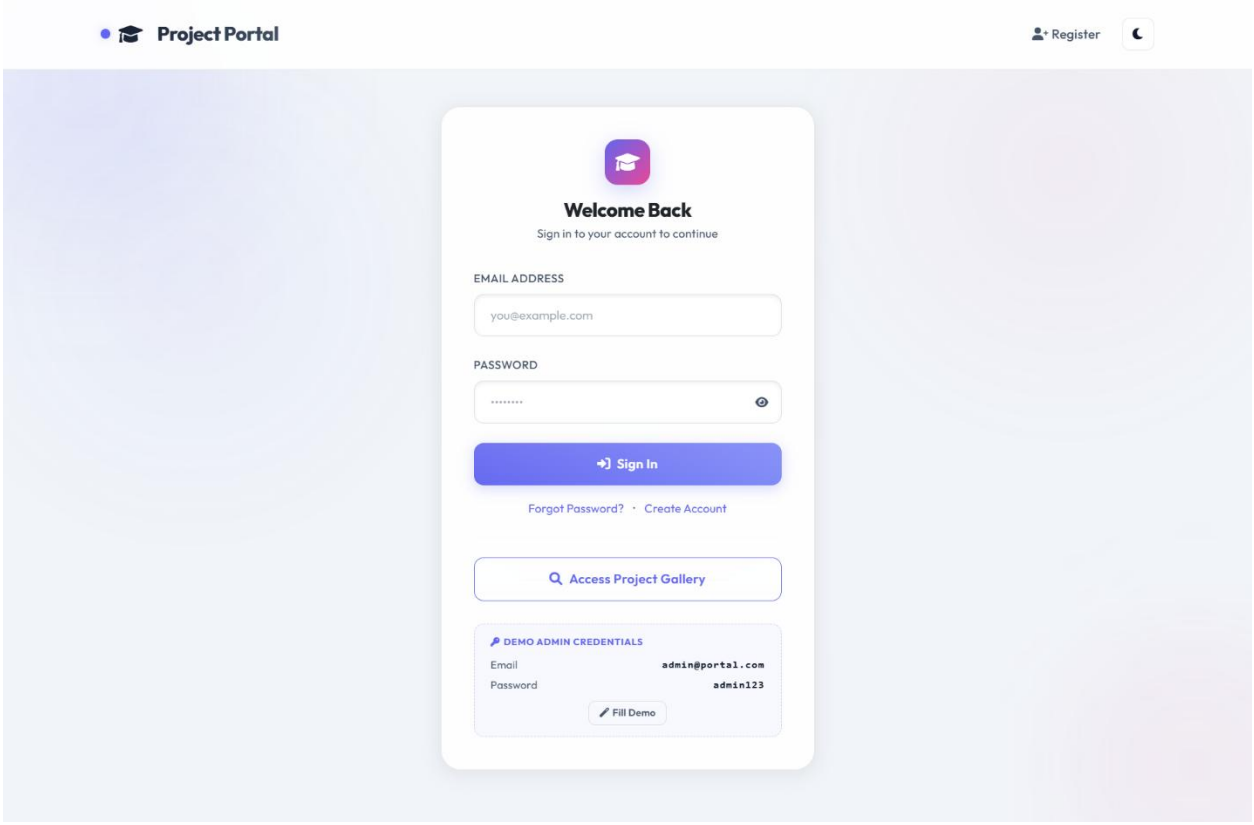


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

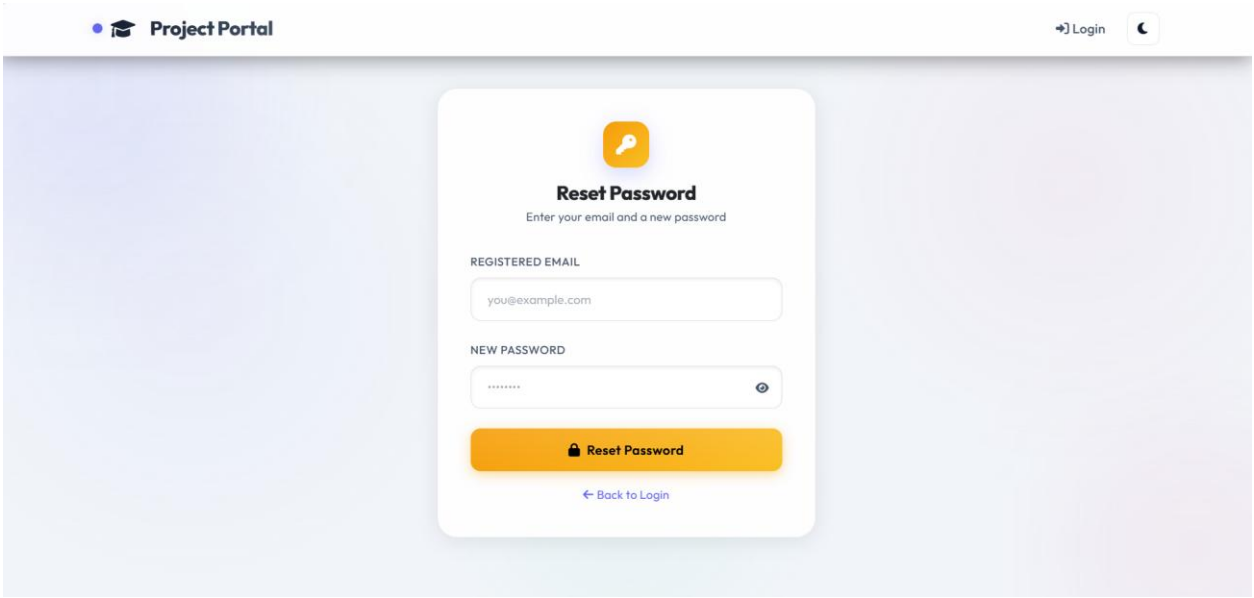
Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
role_id	int	NO	MUL	NULL	
username	varchar(100)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
password_hash	varchar(255)	NO		NULL	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

SCREENS AND DESIGN ELEMENTS

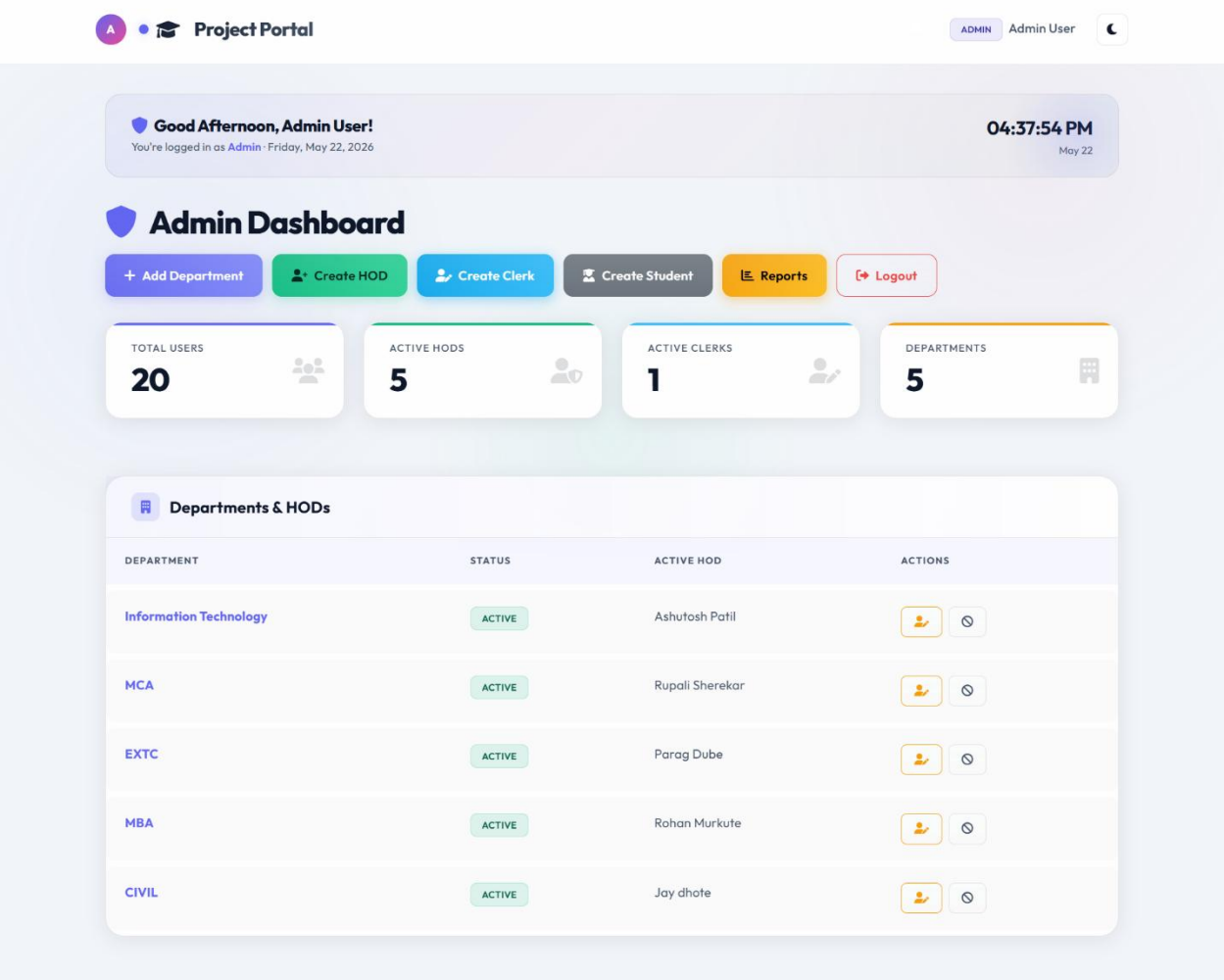
1. Login page



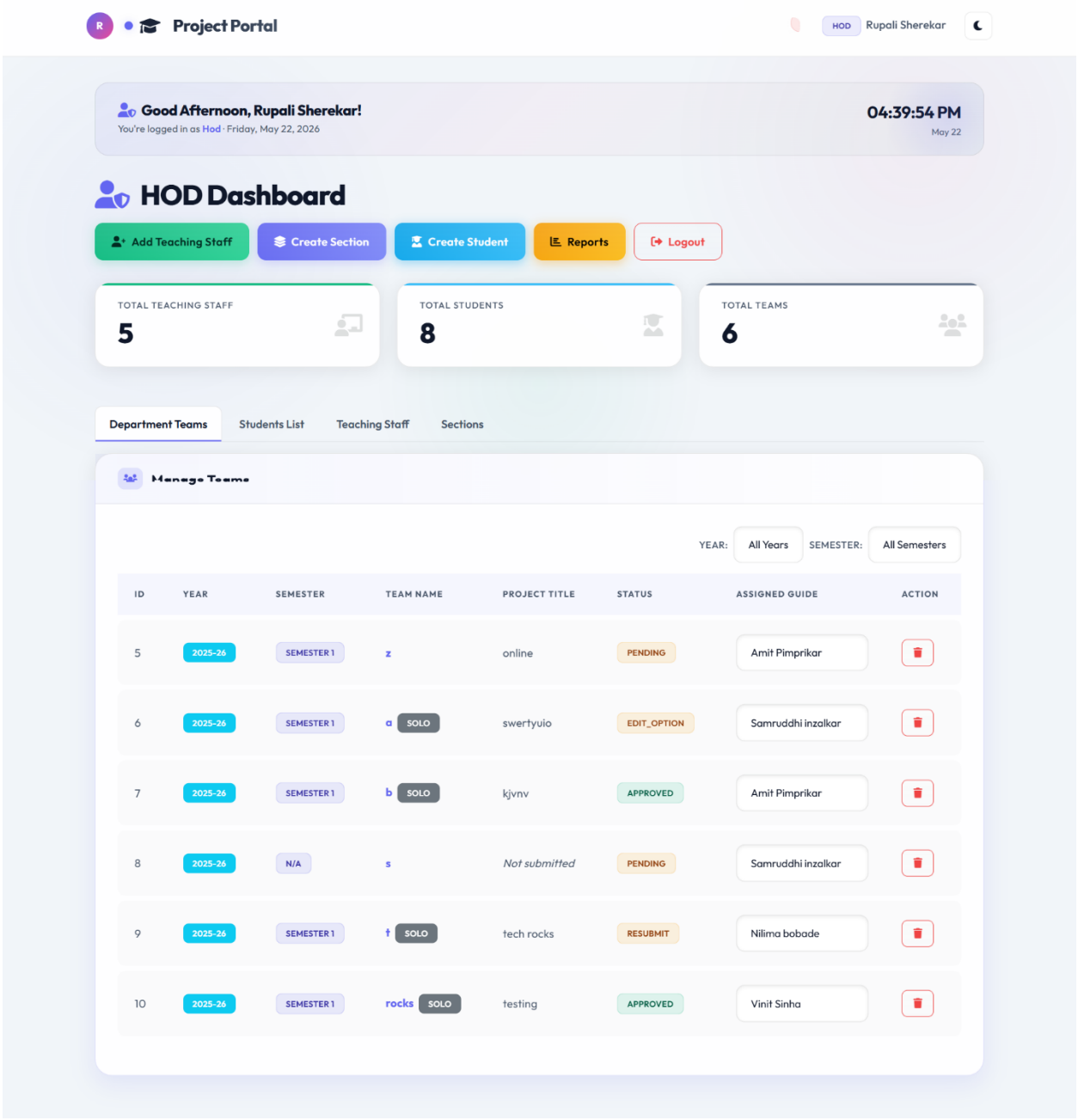
2. Forgot password Page



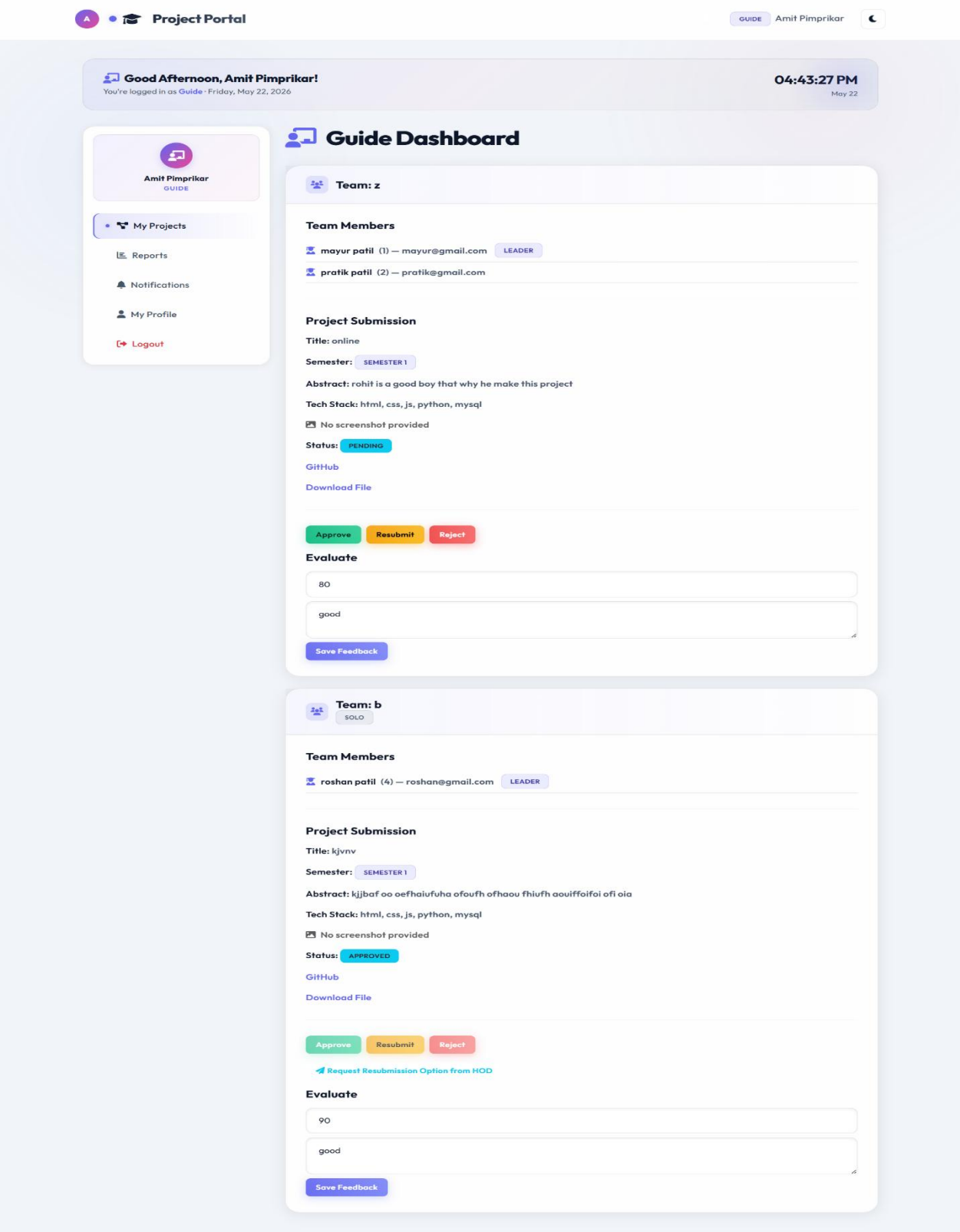
3. After Login as a Admin [Admin Dashboard]



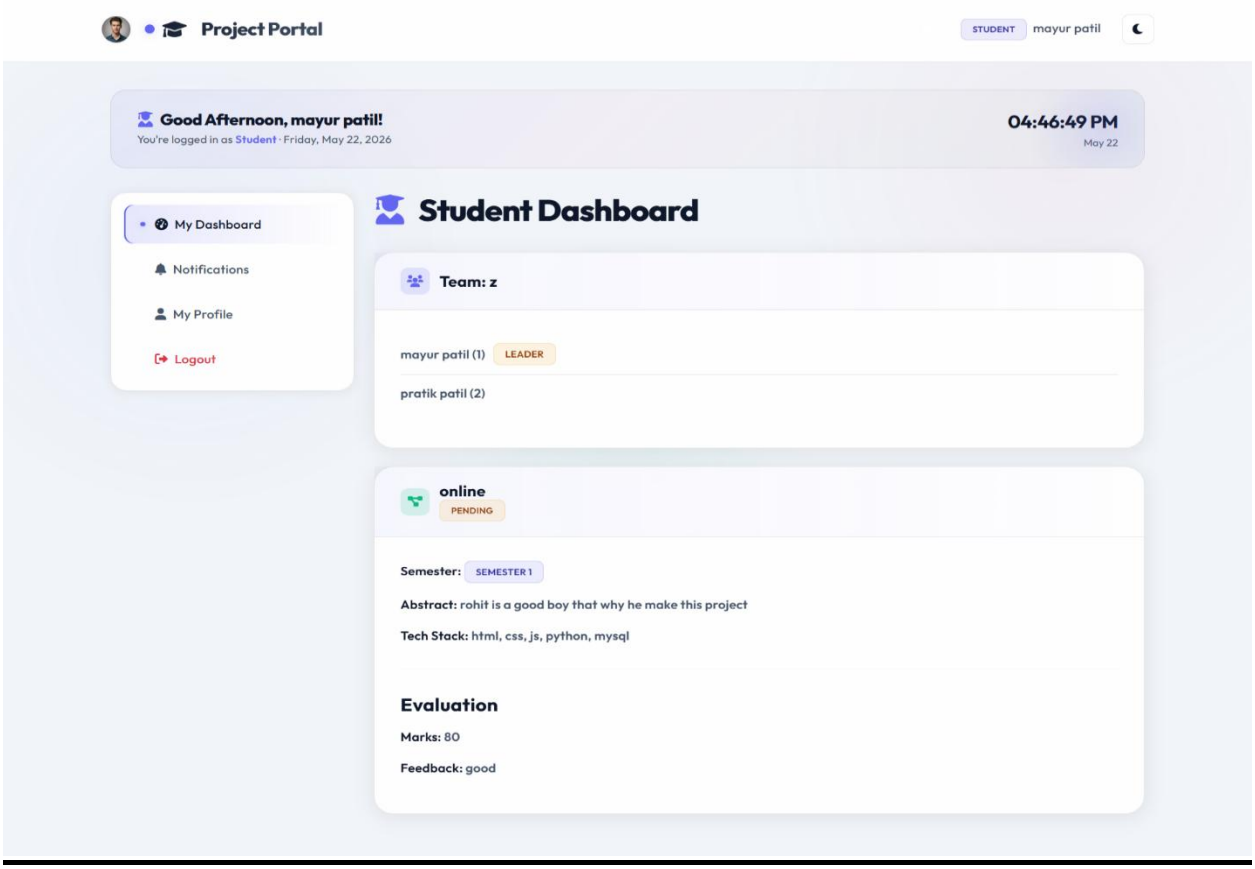
4. After Login as a HOD [HODs Dashboard]



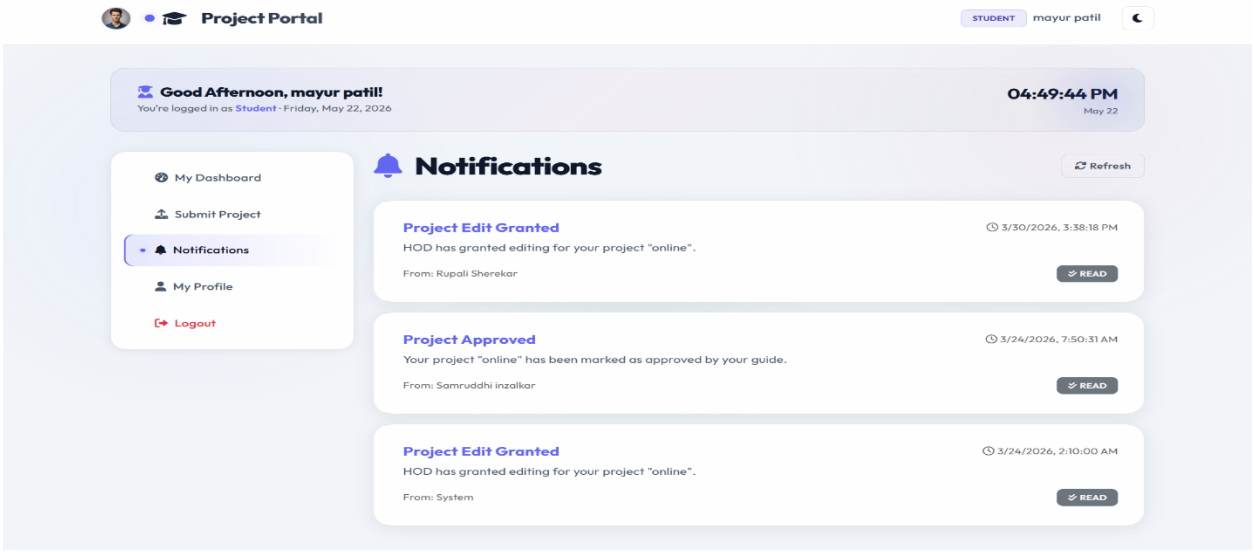
3. After Login as a Guide [Guide Dashboard]



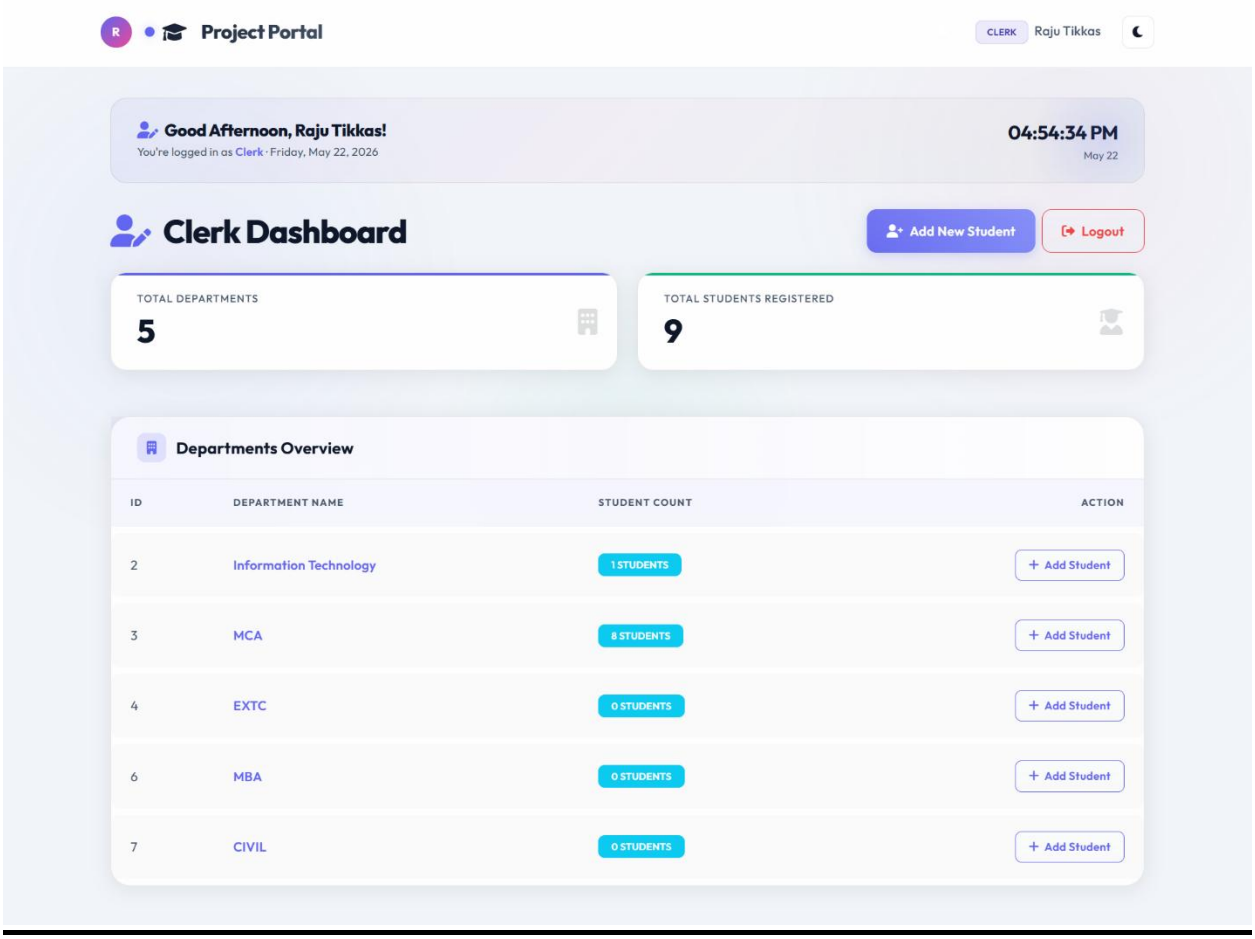
6. After Login as a Student [Student Dashboard]



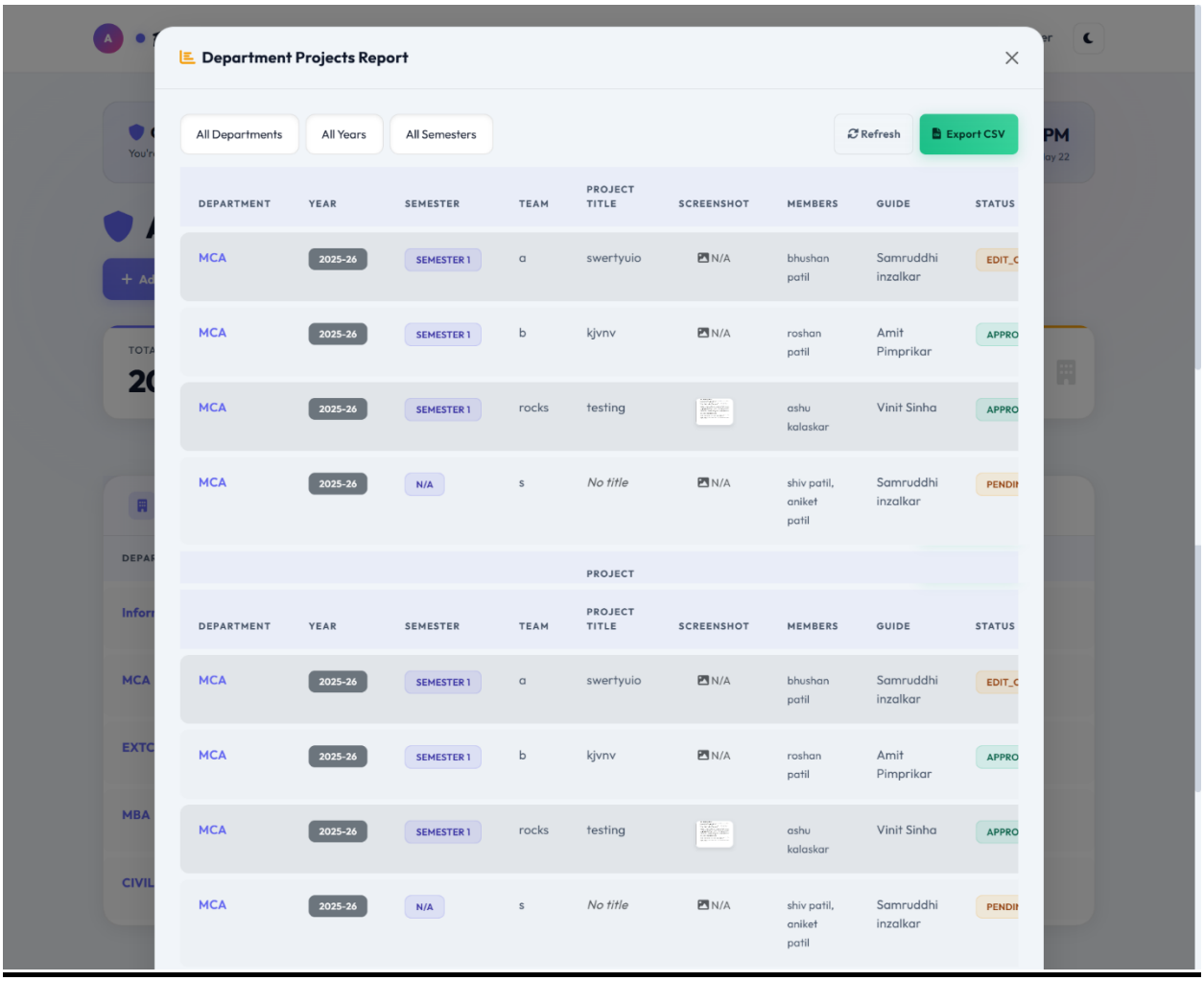
7. Notifications Page



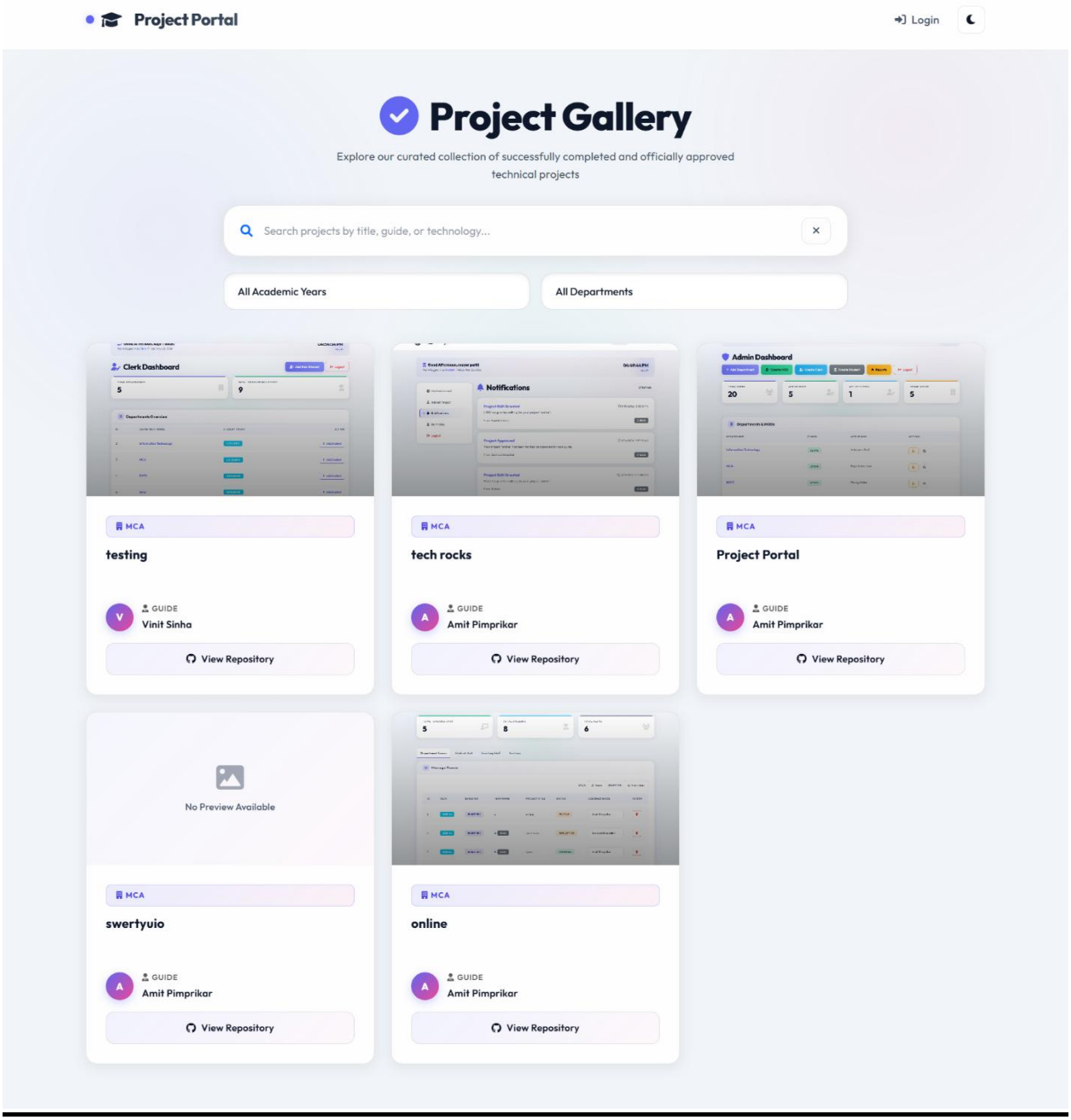
8. After login as Clerk [Clerk Dashboard]



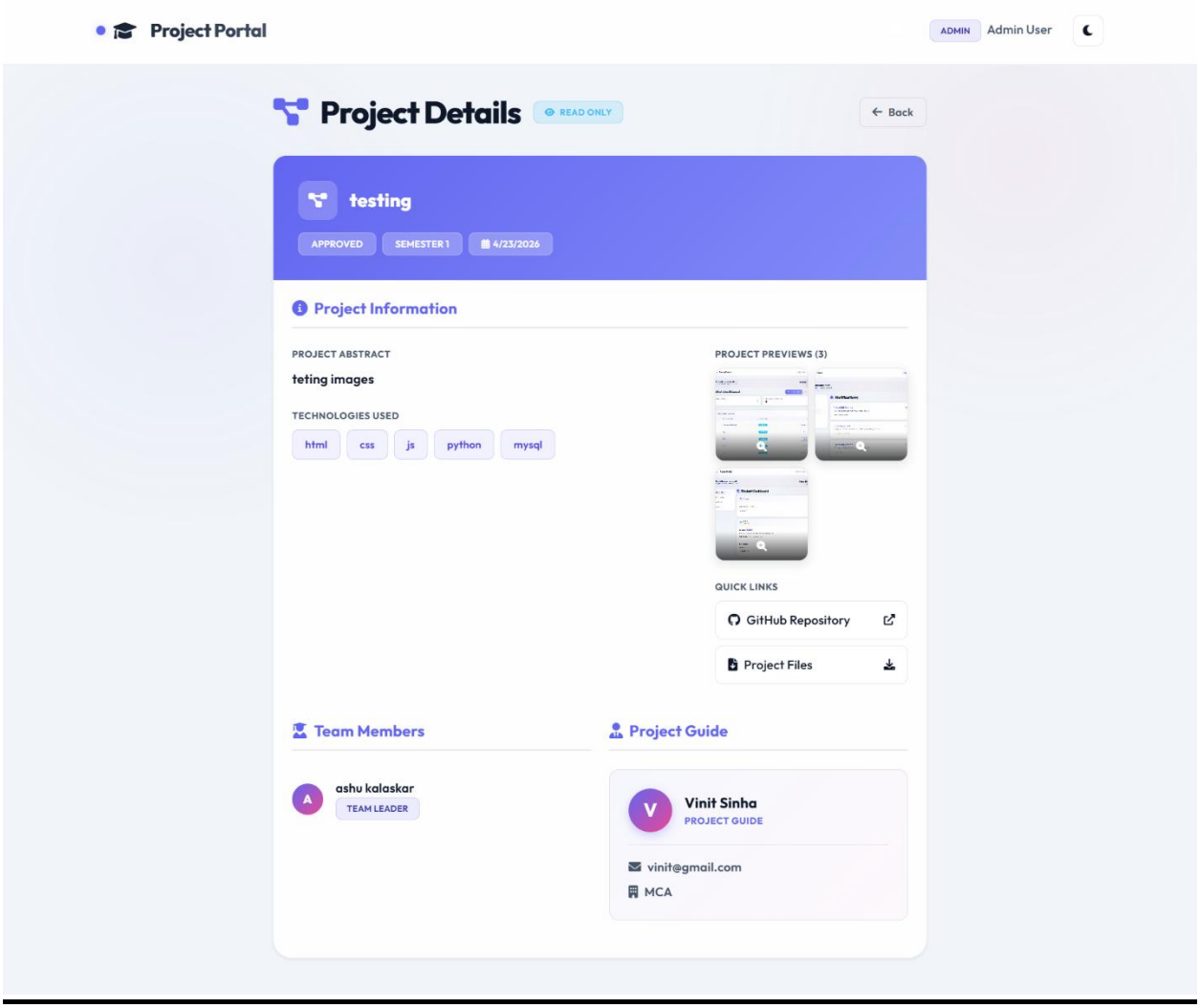
9. Generate Report Functionality



10. Project Gallary



11. Project Details



LIMITATIONS

1. Large File Upload Limitations

- Uploading large ZIP files, project reports, and screenshots may slow down the system performance.
- Users with slow internet connections may experience upload failures or delays.

2. Dependence on Internet Connectivity

- The portal requires a stable internet connection for project submission, review, and dashboard access.
- Poor network connectivity can interrupt uploads and affect user experience.

3. Manual Account Activation

- Some user accounts require manual approval or activation by the administrator.
- Delays in account verification may temporarily restrict access to portal features.

4. Limited Scalability

- The system may face performance issues when handling a very large number of users and submissions simultaneously.
- Additional server resources and optimization may be required for large institutions.

5. No Live Project Execution Support

- The portal supports project submission and review only.
- It does not provide real-time project execution, code compilation, or testing functionality.

6. Storage Management Challenges

- Continuous uploading of project files and reports increases database and server storage usage.
- Regular maintenance and backup management are required to handle large amounts of data.

7. Security and Data Protection Risks

- Since project data and user information are stored digitally, strong security measures are necessary.
- Unauthorized access or security breaches may affect confidential academic data.

8. Limited Technical Knowledge of Users

- Some students or faculty members may initially face difficulty understanding portal functionalities.
- Basic training or user guidance may be required for effective usage.

9. Browser and Device Compatibility Issues

- Certain features may not work properly on outdated browsers or older devices.
- Performance may vary depending on the user's device configuration.

10. Maintenance and Technical Support Requirements

- The portal requires regular updates, bug fixes, and technical maintenance.
- Server downtime or technical issues may temporarily affect system availability.

CONCLUSION

The Online Project Submission Portal successfully addresses the challenges of traditional manual project management systems used in educational institutions. The system provides a secure, organized, and centralized platform for project submission, evaluation, tracking, and storage.

The portal significantly reduces paperwork and manual effort while improving the overall efficiency of project handling processes. It enables smooth communication and interaction between students, guides, and administrators through a user-friendly digital platform.

The system also helps maintain a structured repository of academic projects, reports, and related documents that can be accessed for future reference and learning purposes. Features such as secure authentication, file uploads, progress tracking, and review management contribute to better academic administration.

Overall, the project demonstrates the importance of digital transformation in educational environments by simplifying project management, improving accessibility, and enhancing productivity for both students and faculty members.

FUTURE SCOPE

Future Enhancements:

The Online Project Submission Portal can be further improved by adding advanced features and modern technologies in future versions. Some possible future enhancements are:

- Integration of plagiarism detection tools for automatic verification of project originality.
- Implementation of an AI-powered chatbot to provide instant support and guidance to students and faculty members.
- Calendar and email notification integration for submission reminders, review schedules, and important updates.
- Support for live project demonstrations and online project presentation sessions.
- Cloud storage integration for secure and scalable file management.
- Development of a mobile application for easier access through smartphones and tablets.
- Advanced analytics and reporting features for monitoring project activities, submission trends, and user performance.
- Real-time project status tracking and notification system.
- Integration with university ERP or academic management systems.
- Multi-language support for better accessibility and usability.
- Automated backup and recovery system for improved data protection.
- Role-based access enhancements with advanced security mechanisms.
- Online code execution and testing support for programming-based projects.
- Dashboard customization options for students, guides, and administrators.
- AI-based project recommendation and evaluation assistance system.

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